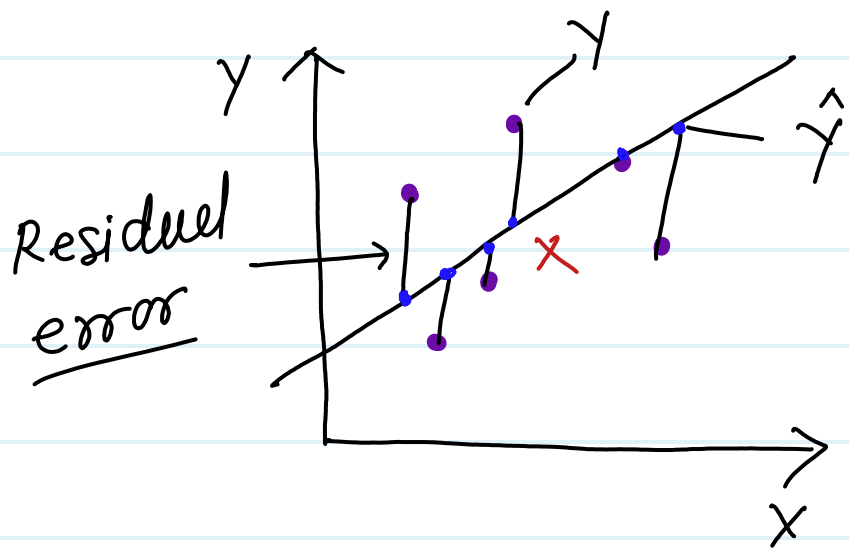
Linear Regression

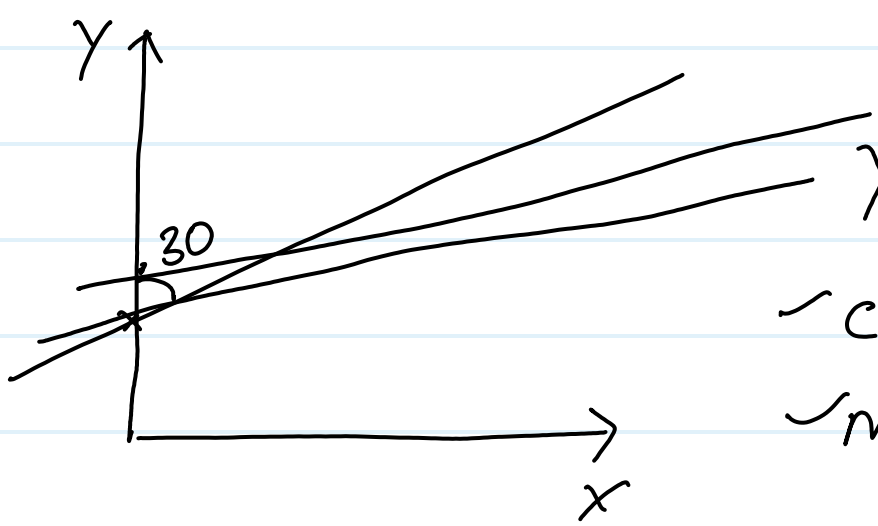
Line eqn $\quad y = mx + c$

y = Actual point

$\hat{y}$ = predicted



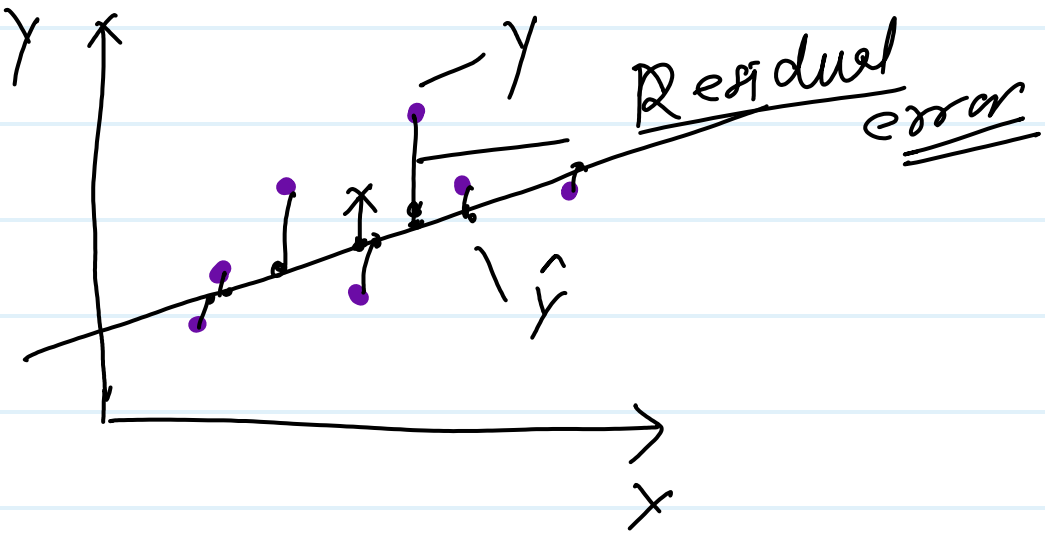
$RE = y - \hat{y}$



$y = mx + c$

- c = intercept
- m

$$\boxed{y = mx + c}$$



Linear Regression Assumption -

1) Data linear
2) No multicolinearity
3) Homoscadacity
4) Distribution of error should be normalized.

Linear Regression

X (hours)	Y (score)
1	50
2	55
3	65
4	70
5	80
6	?

$$y = mx + c$$

$$y = b_1 x + b_0$$

$$y = b_0 + b_1 x$$ – Simple LR.

$$y = b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 + \cdots + b_n x_n$$

multi Linear Reg.

b_1 = slop

b_0 = intercept

mean

$\bar{X} = 3$

$\bar{Y} = 64$

calculation table

X	Y	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X-\bar{X})(Y-\bar{Y})$	$(X-\bar{X})^2$
1	50	-2	-14	28	4
2	55	-1	-9	9	1
3	65	0	1	0	0
4	70	1	6	6	1
5	80	2	16	32	4

$\sum (X-\bar{X})(Y-\bar{Y})$

$\Rightarrow 75$

$\sum (X-\bar{X})^2$

$\Rightarrow 10$

slop calculation

$$b_1 = \frac{\sum (X-\bar{X})(Y-\bar{Y})}{\sum (X-\bar{X})^2}$$

$$b_1 = \frac{75}{10} = 7.5$$

intercept

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_0 = 64 - (7.5 \times 3)$$

$$b_0 = 41.5$$

final Linear Regression

$$y = 41.5 + 7.5.x$$

prediction of eqn

$$x = 6$$

$$y = 41.5 + (7.5 \times 6)$$

$$y = 86.5$$

Prediction table

X	Y	$\hat{Y}$
1	50	49
2	55	56.5
3	65	64
4	70	71.5
5	80	79

Residual error

Residual = Actual − Predicted

$X = 1$

$Y = 50$

$\hat{Y} = 49$

$Y - \hat{Y} = 1$

intution Linear Reg. Least square method

$$\Sigma (Y - \hat{Y})^2$$

X	Y	$\hat{Y}$	$(Y-\hat{Y})$
1	50	49	1
2	55	56.5	-1.5
3	65	64	1
4	70	71.5	-1.5
5	80	79	1

MAE (mean Absolute error)

$$MAE = \frac{\sum |Y - \hat{Y}|}{n}$$

Error	Absolute
1	1
-1.5	1.5
1	1
-1.5	1.5
1	1

$$\Rightarrow \frac{6}{5} = 1.2$$ Avg error

MSE (mean squared error)

$$MSE = \frac{\sum (y - \hat{y})^2}{n}$$

$(y-\hat{y})$	square
-1.5	2.25
-1.5	2.25

$$\Rightarrow \frac{7.5}{5} = 1.5 \quad \text{error}$$

RMSE (Root mean squared error)

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{1.5}$$

$$RMSE = 1.22 \quad \text{Error}$$

R^2 score -

$$R^2 = 1 - \frac{SS_{residual}}{SS_{Total}}$$

SS_{res} = residual sum of square

SS_{total} = total sum of square

$\bar{Y} = 64$ ✓

SS_{res}

$$= \sum (Y - \hat{Y})^2$$

$$= 7.5$$

SS_{total}

$$= \sum (Y - \bar{Y})^2$$

y	$y-\bar{y}$	$(y-\bar{y})^2$
50	-14	196
55	-9	81
65	1	1
70	6	36
80	16	256

$\Rightarrow$ 570

R^2 find

$$R^2 = 1 - \frac{7.5}{570}$$

$$= 1 - 0.013$$

$$R^2 = 0.987$$

$$= 98.7\%$$

LR

$$Y = mx + C$$

$$Y = \theta_1 x + \theta_0$$

$$Y = \theta_0 + \theta_1 x_1$$

$$Y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

θ_1 = slop

θ_0 = intercept

Y = Actual

$\hat{Y}$ = predicted

$Y - \hat{Y}$ = Residual

$$Y = J = h_\theta(x)$$

$$J = J(\theta_0, \theta_1)$$

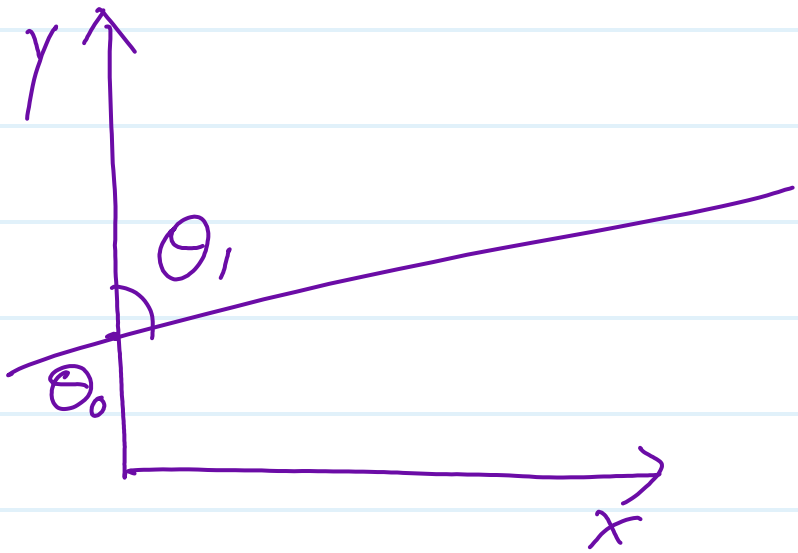
Loss function

$$J(\theta_0, \theta_1) = (y - \hat{y})^2$$

x	y	$\hat{y}$	$y - \hat{y}$
1	50	49	1

cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left[h_\theta(x_i) - (y_i) \right]^2$$



Repeat conversion theorem

$$\theta_j = \theta_j - \alpha \frac{\partial J(\theta)}{\partial \theta_j}$$

θ_0 = some value 1

θ_1 = some value 30

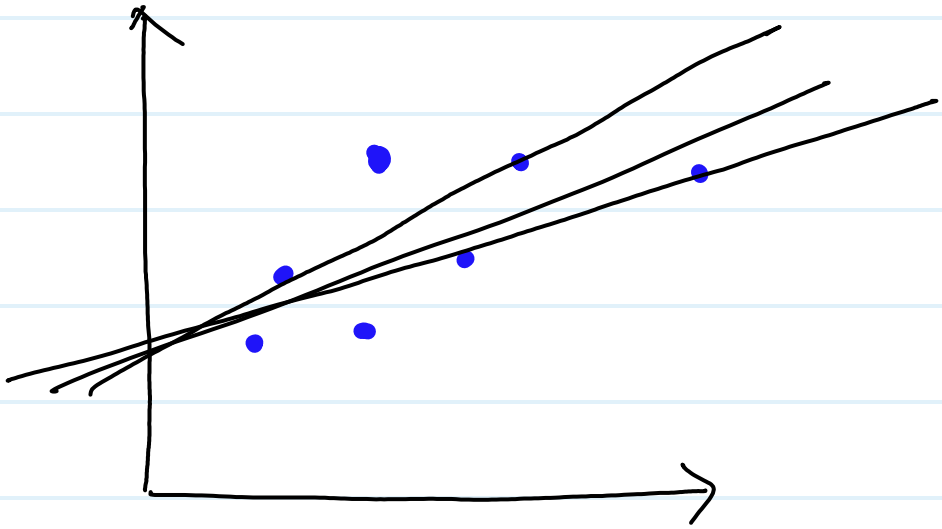
$$\theta_{0_{new}} = \theta_{0_{old}} - \alpha \frac{\partial Loss}{\partial \theta_{0_{old}}}$$

$$\theta_{1_{new}} = \theta_{1_{old}} - \alpha \frac{\partial Loss}{\partial \theta_{1_{old}}}$$

Gradient Decent Algorithm

α = learning rate

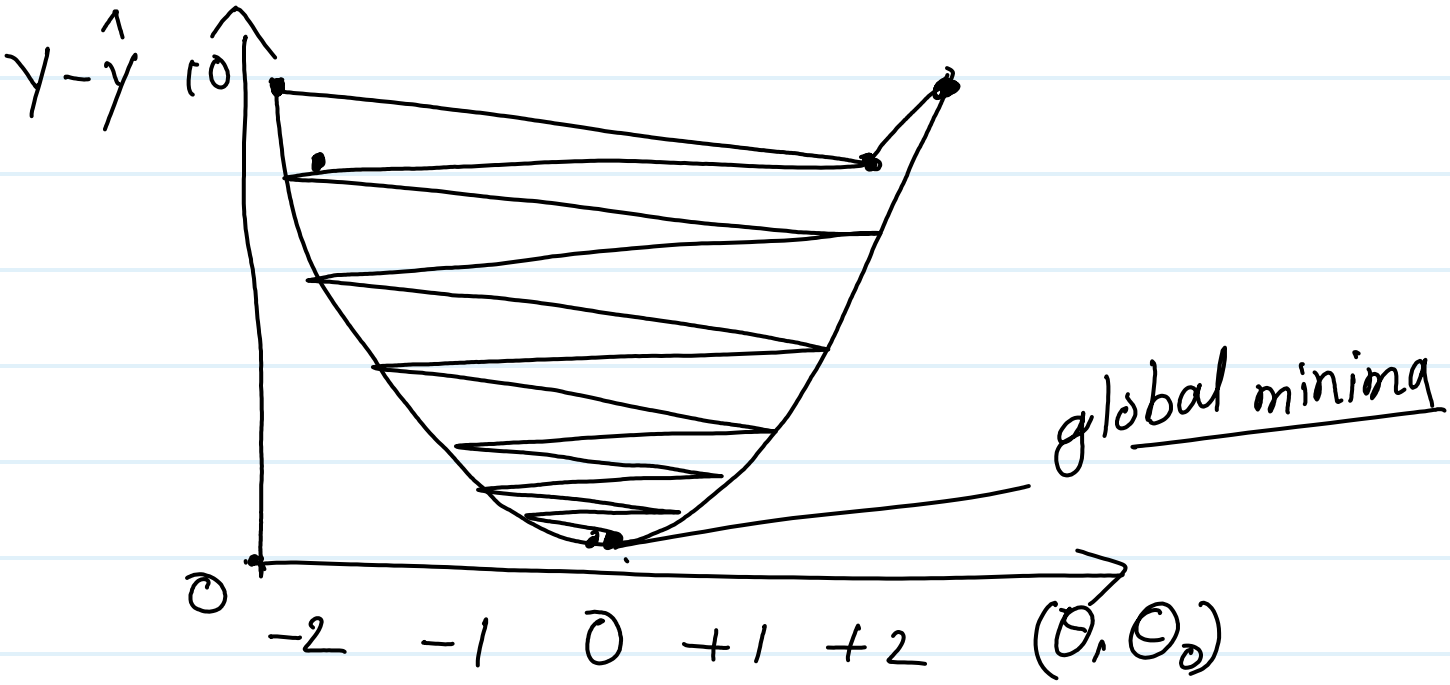
[0.1, 0.01, 0.5, 1]



$\theta_1 = 7.5$

$\theta_0 = 41.5$

Repeat conversion



Adjusted R^2 score –

$$\text{Adj. } R^2 = 1 - \frac{(1-R^2)(N-1)}{N-P-1}$$

N = no. of datapoint in our dataset

P = no. of independent feature